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	Structure determination of large proteins, viruses and protein complexes is accomplished using X-ray diffraction techniques. Exposure of biological crystals to ionizing X-rays, and particularly to high flux third generation synchrotron sources, results in the formation of free radicals, which can alter and damage molecular and crystal structure. Cryogenic cooling of biological crystals has led to significantly longer lifetimes of crystals exposed to the ionizing radiation of X-rays. This improvement of crystal lifetime is largely attributed to the reduced diffusion of free radicals in the completely solidified biological crystal matrix at cryogenic temperatures. Protein crystals contain a significant fraction of solvent in the large channels within the crystal. This solvent is primarily disordered water. In bringing the crystal to low temperatures, this solvent must be vitrified without the formation of ice, which would expand within the solvent channels, destroying the protein crystal structure.		

Task Description:

In general, amorphous solids are formed with rapid cooling. There is still a great deal of controversy on how to properly flash cool biological crystals for X-ray diffraction studies. The general consensus is that ‘faster is better’. However, without good models of the heat transfer process and with little physical property data for both cryoprotectant solutions and crystals, it is difficult to assess various flash cooling techniques. As crystals become larger, the surface to volume ratio of the crystal decreases. This results in reduced area for heat transfer for a given mass of crystal, making flash cooling of large crystals problematic. Crystals grown in microgravity are expected to be of larger size and to have lower mosaicity than terrestrially grown crystals. It is important for the flash cooling technique to be controlled so that mosaicity is not adversely affected and large crystals can be flash cooled without having to resort to techniques such as sectioning which can introduce crystal defects. Any benefits obtained from crystal growth under microgravity conditions can be completely removed by damage induced during flash cooling of crystals. Predictive models of heat transfer will allow optimization of flash cooling protocols for a variety of crystal sizes and geometries. The overall goal of this project is to develop predictive heat transfer models for cooling protein crystals in helium and nitrogen gas streams and to measure some of the physical and transport properties needed for these models. The overall goal of this project is to optimize flash cooling protocols for protein crystals. In order to generalize these protocols for crystals of varying size, using either nitrogen or helium streams at varying flow rates and temperatures a predictive heat transfer model will be developed. The crystal diffraction quality will then be assessed for crystals subjected to varying cooling protocols. This research will focus on the three specific objectives listed below. 1. Develop a predictive model for convective and conductive heat transfer for an open flow helium and nitrogen cryostat.

2. Measure thermal and physical properties of cryoprotectant solutions and protein crystals at low temperature.

3. Using the predictive heat transfer model and system properties, compare resulting crystal quality for crystals subjected to varying cooling protocols. Protein crystallization is the essential first step in X-ray crystallographic structure determination of biological macromolecules. For these studies, protein crystal quality and crystallinity are very important. There is strong evidence that protein crystals grown in microgravity have reduced mosaicity and are larger in size than terrestrially grown crystals. In order to maintain the protein crystal quality, structure and composition of both microgravity grown and terrestrially grown crystals in ionizing X-ray radiation, biological crystals are cooled to cryogenic temperatures. This flash cooing process is more difficult with larger crystals which have smaller surface to volume ratios. Cryogenic cooling of biological crystals has led to significantly longer lifetimes of crystals exposed to the ionizing radiation of X-rays. This improvement of crystal lifetime is largely attributed to the reduced diffusion of free radicals formed in the ionizing beam. Optimization of the flash cooling process will prevent loss of biological crystals during cooling. This is especially important for biological molecules that are available in limited quantities.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

A cryostat from CryoIndustries was purchased and installed. This cryostat uses either helium or nitrogen as cryogen and can operate over a range of temperatures and flow rates. Calibration of the cryostat is underway and development of a heat transfer model has started. A thermo conductivity meter that measures unsteady state heat conduction through the sample was purchased and calibrated. The instrument will be used to measure thermal conductivity of protein crystals. Trends in radiation damage response and crystal quality between N2 and He flash cooled crystals were compared and results published in The Journal of Synchrotron Research. Data were collected from crystals of xylose isomerase (XI) and the anti-ssDNA antibody Fab (BV0401) at the Advance Photon Source (APS) IMCA-CAT station at Argonne National Laboratory. The composite trends for the two sets of data from XI crystals shown in Table 1 include greater increases in both the unit cell volume and Wilson B metric with increased radiation load (increased time) for nitrogen temperature data collection. For helium data, there is a lower Rmerge and much lower decrease in the signal-to-noise ratio in the highest resolution shells with increased radiation exposure, compared with the data collected from the N2 cooled crystal. Similar comparisons were extracted from diffraction data of crystals of BV0401. The results from the reprocessing of these data are shown in Table 2 below. The data from the two BV0401 crystals were similar in other diffraction measurements. The overall Rmerge was 4.97% for the N2 data and 5.21% for the He data; the overall I/sI ratios were also similar. The BV0401 results reinforce the XI observations: the overall trend suggests that crystals maintain order, and high resolution data are less affected by increased radiation load when crystals are cooled with He rather than N2.

Vitrification of Cryoprotectant Solutions. In order to assess the impact of sample size on flash cooling, a series of Hampton Screen I crystallization solutions were flash cooled to ascertain the minimum glycerol concentration needed for vitrification of solutions in loops of nominal sizes of 1.0, 0.5 and 0.1 mm. A 100 K nitrogen gas stream was generated using an Oxford Cryostream series 600 nitrogen cryostat. Glycerol concentration was varied in 5 % (v/v) increments. Glassy solutions appear visually transparent (figure 1). X-ray data were collected with 30 to 90 second frames with the loop perpendicular to the primary beam using a Bruker 6000 CCD diffractometer. X-ray data were integrated as a powder diffraction pattern using GADDS software (Bruker) and plotted as intensity, I, versus diffraction angle, 2 theta. These plots were closely examined in the region of expected ice diffraction. For vitrified samples, I versus 2 theta plots exhibit diffuse scatter and sharp ice diffraction peaks are absent from the diffraction pattern. These plots were closely examined for glycerol concentrations 5 to 10 % above and below the point where distinct ice diffraction disappeared. Baseline intensity was calculated using a polynomial least squares regression of I versus 2 theta data before and after the region of ice diffraction. The sample intensity at the most intense ice peak position (2 theta of 10.8°, corresponding to a d-spacing of 3.74 Å) was compared with background intensity at that position calculated from the polynomial expression. Samples were considered to be vitrified when the ratio of sample intensity to background was less than 1.35. The glycerol concentration needed for vitrification was found to be dramatically lower for the smaller loop sizes. Indeed, if the sample is small enough, no cryoprotectant is required.

Glass transition temperatures of solutions used to vitrify samples in the 1 mm cryoloop were measured and found to fall within a narrow range of values (data not shown). The Tg required to vitrify samples in the 1 mm loop is consistently between about 157 and 159 K for ten of the twelve Hampton screen solutions examined. Screen number 7 requires a Tg less than or equal to 160 K and screen 32 requires a Tg less than 157 K. There is an uncertainty of about 1 to 2 K in Tg values since the glass transition is a kinetic phenomenon (not a thermodynamic property). The twelve samples vitrified in the 1 mm cryoloops required approximately the same Tg values (within the level of uncertainty in this value). Since cooling rates for samples in 1 mm cryoloops are expected to be invariant provided that sample thermal and transport

	properties have similar values and temperature dependency, this may point to a ‘critical vitrification time’ in which sample temperatures can fall between the melting point and glass transition without ice formation. Smaller samples require less cryoprotectant which widens the temperature difference between melting and glass transition temperatures, but these samples cool more rapidly than larger samples, possibly leading to similar times between melting and glass transition for all samples (this time may also be scaled to sample size). The sample temperature history can be calculated provided temperature dependent thermodynamic and transport properties are known. These measurements are currently underway.
Bibliography Type:	Description: (Last Updated: 10/11/2007)
Articles in Peer-reviewed Journals	Hanson, B.L.; Schall, C.A.; Bunick, G.J. "New techniques in macromolecular cryscrystallography: Macromolecular crystal annealing and cryogenic helium." J. Struct. Biol. 2003 Oct;142: 77-87. , Jan-2004
Presentation	Chinte, U.; Shah, B; Hanson, B.L.;Chen, Y.-S.; Kirschbaum, K.; Pinkerton, A.A.; DeWitt, K.; Schall, C. "Comparison of glycerol concentration required to vitrify Hampton Screen 1 solutions using helium or nitrogen as the cryogen" American Crystallographic Association Annual Meeting Jul-2003